AgriFlow Foods

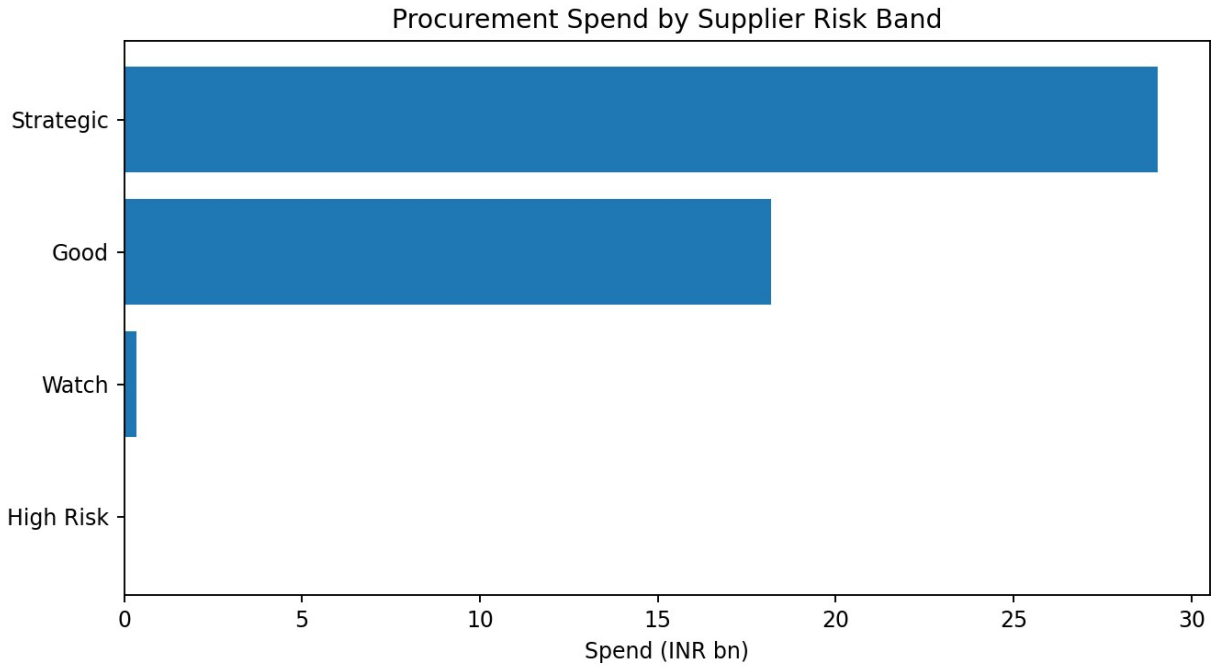
Agricultural Supply Chain Analytics — Instructor Solution Report

1. Executive Baseline

Metric	Reference Result
Procurement spend	INR 47.6 bn
Net sales in customer-order feed	INR 7.5 bn
Average fill rate	94.1%
Strict project OTIF	10.9%
Quality rejection rate	6.2%
Transport loss rate	5.8%
Temperature-excursion rate	52.9%
Current spoilage value	INR 578.8 m
Forecast holdout WAPE	10.4%
Simulated recall lot	LOT0005168
Recall-affected customers	19

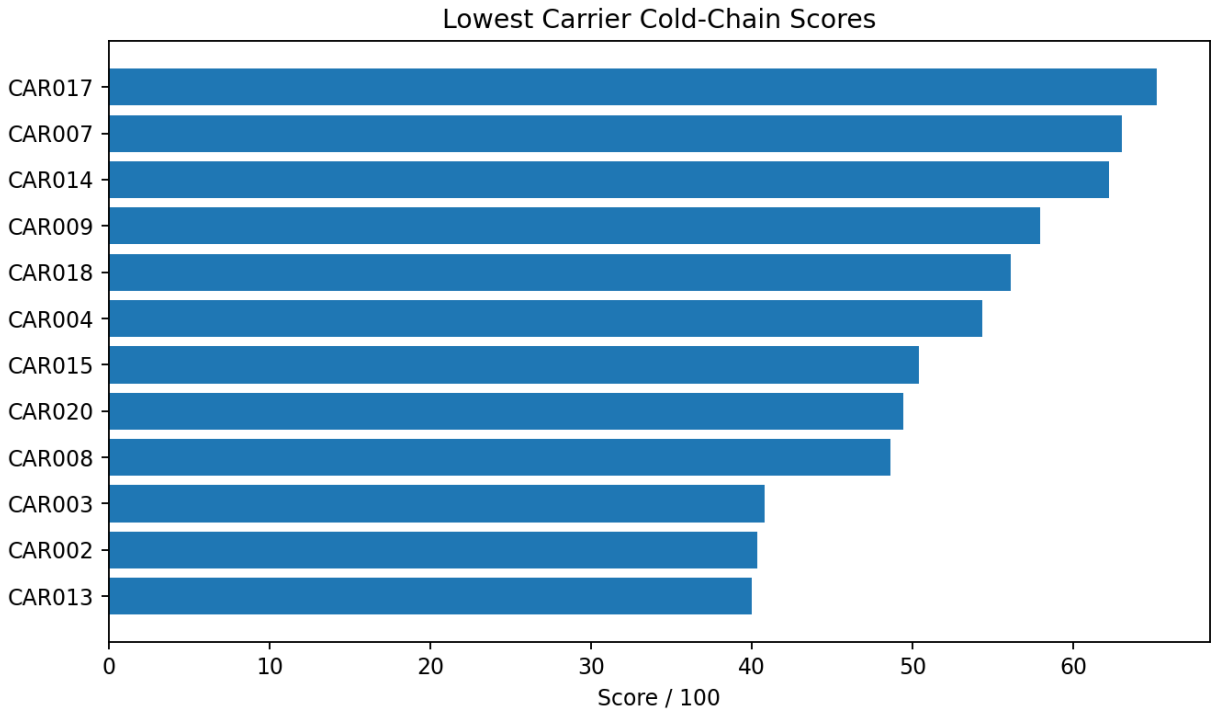
The OTIF definition in this training case is deliberately strict (in-full and on/before promised date), creating a visibly underperforming baseline that learners must diagnose rather than normalize away.

2. Supplier / Procurement Diagnosis



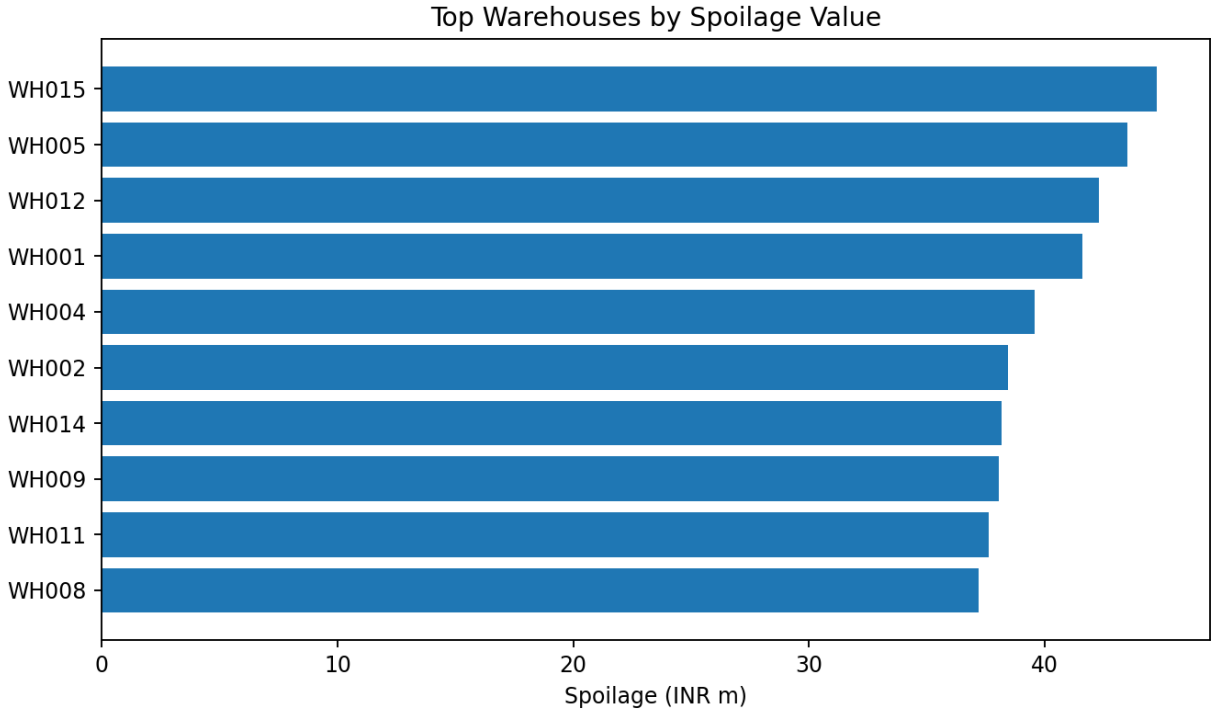
Supplier ranking should combine yield attainment, quality acceptance, compliance and procurement price variance. A low farm-gate price is not necessarily a good sourcing decision if it increases reject, delay or loss downstream.

3. Cold Chain & Logistics



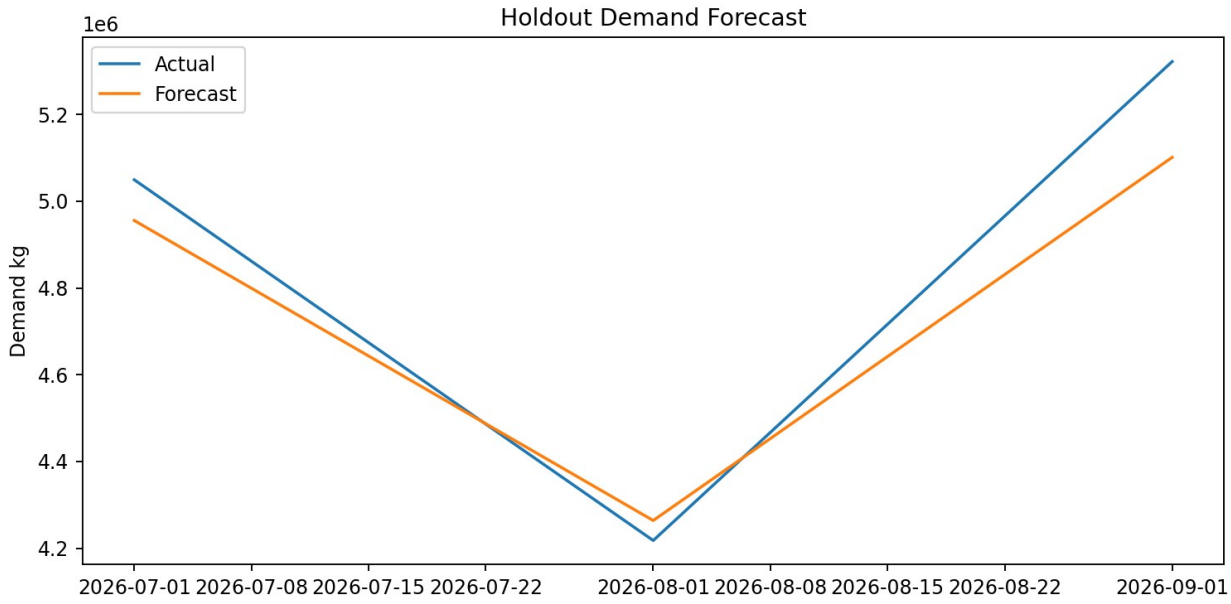
Approximately 52.9% of inbound shipments record more than two hours of temperature excursion in the synthetic baseline. The appropriate response is lane/carrier/product targeting, not blanket cold-chain capex.

4. Warehouse / Shelf-Life / Spoilage



The current residual-stock spoilage value is approximately INR 578.8m. Use warehouse, product, remaining shelf life and temperature history to isolate where FEFO, planning or cold-storage controls have the greatest value.

5. Demand Forecasting & Inventory Policy



The reference lag-based Random Forest produces a holdout WAPE of 10.4%. Learners should then translate forecast/demand variability and average lead time into product-specific safety stock and reorder points, while considering shelf life for perishable items.

6. Traceability / Recall

Field	Reference
Suspect lot	LOT0005168
Affected orders	20
Affected customers	19
Allocated quantity kg	65,071
Sales exposure INR	2,909,163

The recall exercise is a key differentiator: the learner must start from one lot, identify its farm and warehouse path, and enumerate every downstream order/customer. In a real food business, this is both an analytics and governance requirement.

7. Financial Opportunity Sizing

Initiative	Addressable Pool	Low	Base	High
Cold-chain excursion reduction	INR 420.9m	8%	18%	30%
Supplier quality improvement	INR 3,052.0m	6%	12%	20%
Fill-rate recovery	INR 440.8m	10%	25%	40%
Inventory expiry reduction	INR 578.8m	5%	12%	22%

The reference base-case capture assumptions imply about INR 621.6m of potential impact. This is a scenario, not a commitment; real investment cost, implementation constraints and causal validation would be required.

8. Reference 90-Day Management Agenda

Priority	Action	Primary KPI
1	Carrier/lane cold-chain control tower for high-perishability products	Excursion rate; transport loss
2	Supplier quality improvement / reallocation for high-spend low-score farms	Quality rejection; supplier score
3	Shelf-life & FEFO exception queue at high-loss warehouses	Spoilage value; critical shelf-life inventory
4	Product-level S&OP reset using forecast WAPE, lead time and shelf life	Fill rate; safety stock; expiry
5	Lot-level recall playbook and traceability dashboard	Trace completion time; affected orders identified

9. Grading Guidance

- Reward correct physical-flow reasoning, not just technically valid code.
- Require reconciliation of kilograms and INR before recommendations.
- Look for total-delivered-cost thinking rather than cheapest-source logic.
- Require forecast validation on a holdout period, not in-sample fit.
- Require explicit assumptions in safety stock and opportunity sizing.
- The final answer should be a short management agenda supported by evidence, not a tour of every chart.